

Exp.
06

Wind Tunnel Experiment: Measurement of Airfoil Drag Coefficient

Objective: To determine the drag coefficient (C_d) of an airfoil using a subsonic wind tunnel by measuring the aerodynamic drag at different free-stream velocities and comparing the results with theoretical predictions.

Relevant theoretical background:

We will test airfoil in this experiment. And the obvious question that arises is: what is an airfoil?

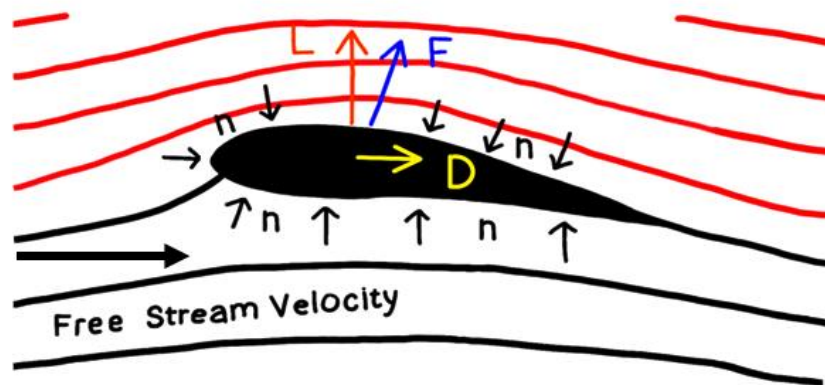


Figure 1: Schematic of an airfoil and the force components acting on it (image source: Glenn research center)

An airfoil is a specially shaped body designed to generate lift when air flows over its surface. The front-most point of the airfoil is called the leading edge, while the rear-most point is known as the trailing edge. The straight line joining these two points is called the chord, and its length is referred to as the chord length. The curved upper and lower surfaces guide the airflow around the airfoil and produce a pressure difference that generates lift. The orientation of the airfoil relative to the incoming airflow is described by the angle of attack, which is the angle between the chord line and the free-stream velocity. These geometric parameters strongly influence the aerodynamic forces acting on the airfoil.

The forces on an airfoil: The aerodynamic force acting on an object placed in a moving fluid can be resolved into two mutually perpendicular components: **lift**, which acts perpendicular to the incoming flow, and **drag**, which acts parallel to the flow direction. While lift is the desirable force generated by an airfoil, drag represents the aerodynamic resistance opposing motion. The performance of aircraft wings, turbine blades, unmanned aerial vehicles, and many engineering components depends on maximizing lift while minimizing drag.

Unlike bluff bodies, whose drag is primarily dominated by pressure (form) drag due to large-scale flow separation, an airfoil is designed to streamline the flow and delay separation. Consequently, the drag experienced by an airfoil is composed of several mechanisms:

- **Skin-friction drag** is produced by viscous shear stresses within the boundary layer.
- **Pressure (form) drag** arises from pressure differences caused by flow separation.

- **Induced drag** is associated with lift generation on finite-span wings through wingtip vortices (not significant in two-dimensional airfoil testing).
- **Wave drag** occurs only at transonic and supersonic speeds and is absent in this experiment.

For the moderate Reynolds numbers and subsonic speeds encountered in this laboratory, the measured drag is primarily the sum of skin-friction drag and pressure drag.

As air flows over an airfoil, a thin boundary layer develops along its surface. Initially, the boundary layer is typically laminar, producing relatively low skin-friction drag. As the flow proceeds downstream, disturbances amplify and the boundary layer may transition to turbulence. Although a turbulent boundary layer has higher wall shear stress, it possesses greater momentum near the surface and is therefore more resistant to adverse pressure gradients. Consequently, turbulent flow often delays separation and reduces pressure drag. The balance between these competing effects determines the total drag of the airfoil.

For streamlined airfoils operating at small angles of attack, skin-friction drag dominates. When the angle of attack increases or flow separation occurs, pressure drag dominates. The drag coefficient is a dimensionless parameter that compares the measured drag force with the dynamic pressure of the flow:

$$C_D = \frac{F_D}{\frac{1}{2}\rho V^2 A} \dots\dots\dots (1)$$

Where, D is drag force (N), ρ is air density (kg/m^3), V is free-stream velocity (m/s), and A is the reference area (typically projected/frontal area or planform area as specified by the experiment). The drag coefficient allows comparison of aerodynamic performance independent of speed and size.

To measure the drag force acting on an object, a wind tunnel will be used. The streamlined test object will be mounted inside the test section, and air will be passed over it while the aerodynamic forces acting on the object are measured.

Wind tunnel: A wind tunnel is an experimental facility used to study the behavior of air flowing over objects under controlled laboratory conditions. Instead of moving the object through the air, the object is held stationary while air is driven past it at a known velocity. This approach allows engineers to investigate the aerodynamic characteristics of aircraft wings, automobiles, bridges, wind turbines, buildings, sports equipment, and many other engineering systems in a safe, repeatable, and cost-effective manner. By accurately controlling the airflow velocity and measuring the forces acting on a test model, valuable information regarding aerodynamic performance can be obtained before full-scale prototypes are manufactured.



Figure 2: A subsonic wind tunnel (image source: Tec equipment)

A typical subsonic wind tunnel consists of several major sections that work together to produce a uniform airflow. Air is first drawn through a settling chamber where screens and honeycombs reduce turbulence and straighten the flow. The air then passes through a contraction section, which increases the flow velocity while further improving flow uniformity. The test section is the region where the airfoil is mounted and aerodynamic measurements are performed. Downstream the test section, the diffuser gradually slows the airflow, recovering pressure before the air finally passes through the fan and exits the tunnel. The quality of the airflow in the test section is essential for obtaining accurate aerodynamic measurements.

Measurement of the flow speed:

Pressure of a flowing fluid can be thought of as a combination of two major constituents, (i) static pressure which measures the potential part of the pressure, this is momentum transfer from the random molecular motion in the fluid. Even if the fluid is not moving, it still imparts momentum to the surface. It comes from the internal energy of the fluid. (ii) dynamic pressure comes from the bulk motion of the fluid. It is the kinetic energy of the flow. Mathematically expressed as:

$$P_{\text{total}} = P_{\text{static}} + P_{\text{dynamic}} = P_{\text{static}} + \frac{1}{2} \rho v^2 \dots\dots\dots (2)$$

here ρ is the density of the flowing fluid (in our wind tunnel experiment, this is essentially air), and v is the air speed.

Pitot-static tube for measurement of flow speed:

A pitot static tube utilizes this idea and gives the difference between static and the total pressure, i.e.,

$$P_{\text{dynamic}} = P_{\text{total}} - P_{\text{static}} \dots\dots\dots (3)$$

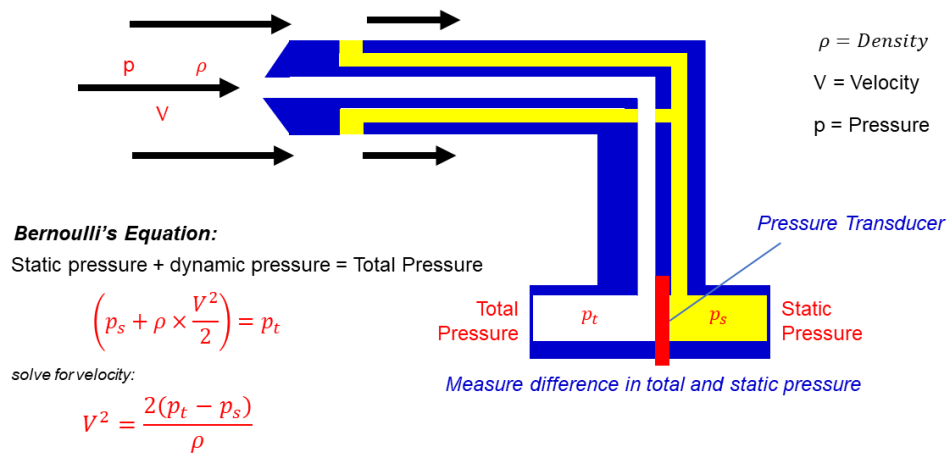


Figure 3: A pitot static tube (image source: Glenn research center)

This makes it possible to compute the velocity as:

$$v = \sqrt{\frac{2P_{\text{dynamic}}}{\rho}} \dots\dots\dots (4)$$

Note, a differential manometer is attached to the pitot-static tube (one measures the stagnation or total pressure, and the other measures the static pressure). The manometric height needs to be converted into pressure as:

$$P = \rho_{\text{mf}}gh \dots\dots\dots (5)$$

here ρ_{mf} is the density of the manometric fluid, and h is the differential height, i.e., the manometer reading. The manometer scale is calibrated for your ease of use, and you can directly input the manometer reading (in mm H₂O) without requiring the need to convert units. From (4) and (5), the velocity can be obtained as:

$$v = \sqrt{\frac{2\Delta P g}{\rho}} \dots\dots\dots (6)$$

Here,

ΔP is in units of mm H₂O. And ρ is air density in kg per cubic meters, you can obtain this value from:

$$\rho = \frac{P_a \times 100}{RT_a} \dots\dots\dots (7)$$

The resultant velocity is in m/s units.

Measurement of drag coefficient:

Now, we can control the pressure far more easily, as such, it is convenient to set the pressure to attain a desired flow-speed. From the equation for velocity, you can find the required differential manometric height to set a desired flow speed.

Desired v	Required h
15.0	
20.0	
25.0	

Now set your experiment (don't worry, we shall show you all these in the laboratory) in such a way that a steady required height is obtained. Repeat each measurement three times and take the average.

Data table: The characteristic diameter of the streamlined object is 50 mm. Take the projected/frontal area $A = \frac{1}{4} \pi D^2 = 0.0019635 \text{ m}^2$.

Angle of attack:

V (m/s)	F_D (N)	F'_D (N)	Net Drag, $F = F_D - F'_D$	Drag coefficient, C_D	Average Drag coefficient, $\overline{C_D}$	Standard deviation
15.0						
20.0						
25.0						

From your experiment, you can readily measure the total drag force, F_D from the load cell. Populate till the third column of the data table. At this point, we need a correction.

Correction for support drag:

The airfoil is mounted on a supporting stem that is also exposed to the airflow. Consequently, the measured force includes not only the drag acting on the airfoil but also the drag acting on the support. To determine the true aerodynamic drag of the airfoil, the drag of the support must first be estimated from the calibration curve provided and then subtracted from the total measured drag.

Interpolate the graph in Figure 4 to find the required F'_D corresponding to your velocity. Subtract this value from F_D : this will give you the true aerodynamic drag in column 4 of the data table. This corrected value, referred to as the **net drag**, F , is subsequently used in calculating the drag coefficient.

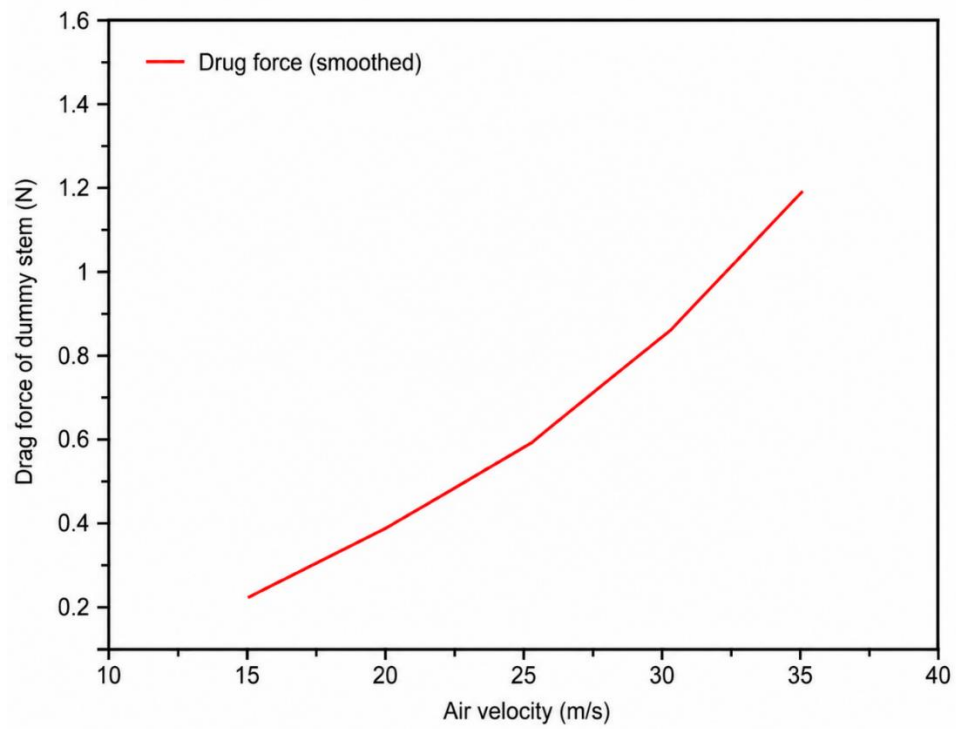


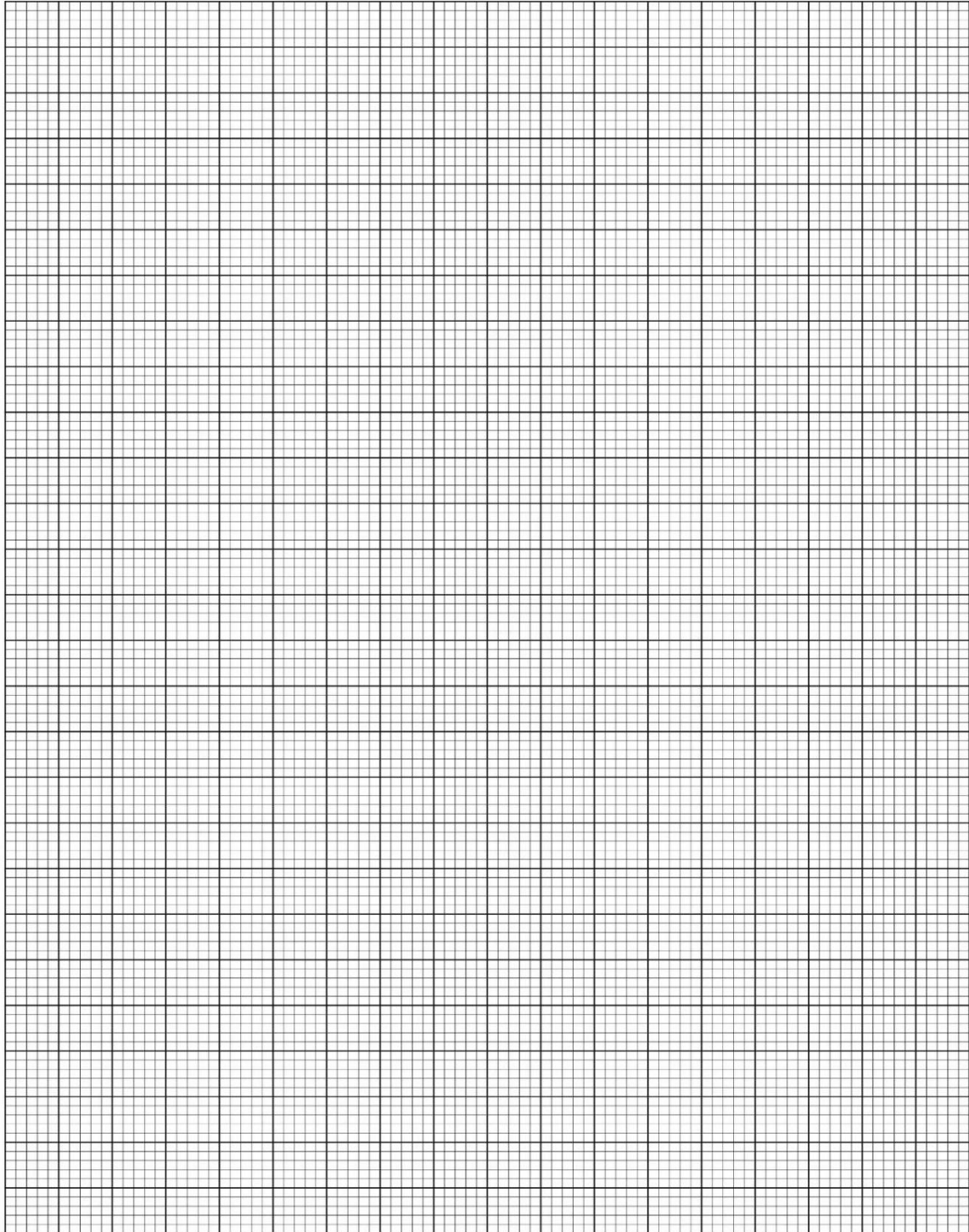
Figure 4: Drag force on the dummy stem (N) for different air velocity

Plot

- From your experimental data plot **coefficient of drag** vs **velocity**. Use standard deviations as your error bars

Along Horizontal Axis:

Along Vertical Axis:



Sample Calculation:

Discussion:

1. What are the sources of uncertainty in your measurement? Discuss.
2. Instead of a suction fan, if a blower were installed at the upstream, would that improve the setup? Discuss.
3. Imagine that you are recruited as the engineer-in-charge for a dedicated wind tunnel testing facility to be developed at IUT. You can choose a lab-space on campus, and IUT gave you this wind tunnel and associated equipment as well as a healthy budget to make any improvement to the setup. What would you do?